

Quiz game Using Python

1. Introduction

The **Quiz Game** is a simple command-line application developed using Python. The program asks the user three general knowledge questions, evaluates each response, and awards one point for every correct answer. At the end of the quiz, it displays the user's final score.

This project demonstrates the use of Python fundamentals such as variables, user input, conditional statements, string manipulation, and the accumulator pattern.

2. Objective

The objectives of this project are:

- Develop an interactive quiz application.
- Accept and process user input.
- Normalize input using string methods.
- Compare user responses accurately.
- Maintain a running score using an accumulator.
- Display the final score using formatted output.

3. Software Requirements

Requirement	Description
Programming Language	Python 3.x
Operating System	Windows, Linux, or macOS
IDE/Text Editor	VS Code, PyCharm, IDLE, Sublime Text, etc.

4. Python Concepts Used

Variables

A variable named **score** is initialized to **0** and acts as an accumulator to keep track of the user's correct answers.

Example:

```
score = 0
```

User Input

The `input()` function is used to capture the user's answer for each question.

Example:

```
answer = input("Enter your answer: ")
```

String Normalization

To prevent incorrect answers caused by extra spaces or uppercase letters, the program uses:

- `.strip()` – Removes leading and trailing spaces.
- `.lower()` – Converts all characters to lowercase.

Example:

```
answer = input("Answer: ").strip().lower()
```

Conditional Statements

The program uses `if-else` statements to determine whether the user's answer is correct.

Example:

```
if answer == "paris":  
    score += 1  
else:  
    print("Wrong Answer!")
```

Accumulator Pattern

The variable `score` stores the cumulative number of correct answers.

- Initial Value:

```
score = 0
```

- Increment:

```
score += 1
```

The score increases **only** when the user provides a correct answer.

Formatted Output (f-Strings)

The final score is displayed using an f-string.

Example:

```
print(f"Your final score is {score}/3.")
```

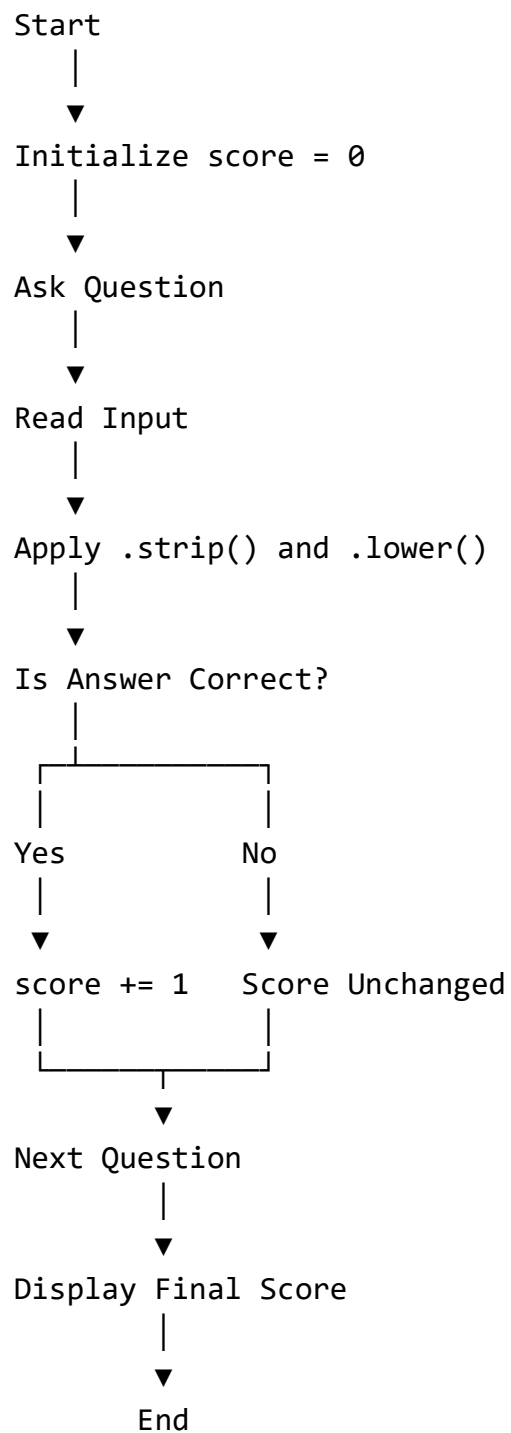
5. Program Workflow

1. Initialize the score to zero.
2. Display the first question.
3. Accept the user's answer.
4. Remove extra spaces using `.strip()`.
5. Convert the answer to lowercase using `.lower()`.
6. Compare the answer with the correct answer.
7. If correct, increment the score.
8. Repeat the same process for all three questions.
9. Display the final score.
10. End the program.

6. Algorithm

1. Start the program.
2. Initialize score = 0.
3. Ask Question 1.
4. Read user input.
5. Apply `.strip()` and `.lower()`.
6. Compare with the correct answer.
7. Increment the score if correct.
8. Repeat Steps 3–7 for the remaining questions.
9. Display the final score.
10. End the program.

7. Logic Flow



8. Features

- Interactive command-line quiz.

- Three predefined questions.
- Case-insensitive answer checking.
- Removes unnecessary whitespace.
- Maintains score using an accumulator.
- Provides immediate feedback after each answer.
- Displays the final score using formatted output.

9. Advantages

- Beginner-friendly implementation.
- Demonstrates essential Python programming concepts.
- Easy to understand and modify.
- Prevents input formatting errors using string normalization.
- Efficient score tracking through an accumulator.

10. Limitations

- Limited to three predefined questions.
- Answers must match the expected text after normalization.
- No database or file storage.
- No randomization of questions.

11. Future Enhancements

- Increase the number of quiz questions.
- Load questions from a file or database.
- Add multiple-choice questions.
- Implement a timer for each question.
- Store high scores.
- Add difficulty levels.
- Randomize question order.

12. Testing

Test Case	Input	Expected Output	Result
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Correct Answer	Paris	Score increases by 1	Pass
Uppercase Input	PARIS	Accepted as correct	Pass
Input with Spaces	Paris	Accepted as correct	Pass
Incorrect Answer	London	Score unchanged	Pass

13. Sample Output

1. What is the capital of France?

Paris

Correct!

2. Which planet is known as the Red Planet?

mars

Correct!

3. What is the largest ocean on Earth?

pacific

Correct!

Quiz Completed!

Your final score is 3/3.

14. Conclusion

The **Quiz Game Using Python** successfully demonstrates fundamental programming concepts such as user input, conditional statements, string normalization using `.strip()` and `.lower()`, and the accumulator pattern for maintaining a running score. The project is simple, interactive, and meets the specified functional requirements while providing a solid foundation for learning Python.